

Import Data

```
In [1]: import pandas as pd
```

```
In [2]: import matplotlib.pyplot as plt
```

```
In [3]: import seaborn as sns
```

```
In [4]: data = pd.read_excel(r"D:\Downloads\Udemy-PY\Sales-Data-Analysis.xlsx")
```

```
In [5]: data
```

Out[5]:

	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7
0	NaN	Order ID	Date	Product	Price	Quantity	Purchase Type	P
1	NaN	10452	2022-11-07 00:00:00	Fries	3.49	573.065903	Online	G
2	NaN	10453	2022-11-07 00:00:00	Beverages	2.95	745.762712	Online	G
3	NaN	10454	2022-11-07 00:00:00	Sides & Other	4.99	200.400802	In-store	G
4	NaN	10455	2022-11-08 00:00:00	Burgers	12.99	569.668976	In-store	
...	...	...	...	...	...	...	...	
250	NaN	10709	2022-12-28 00:00:00	Sides & Other	4.99	200.400802	Drive-thru	G
251	NaN	10710	2022-12-29 00:00:00	Burgers	12.99	754.426482	Drive-thru	G
252	NaN	10711	2022-12-29 00:00:00	Chicken Sandwiches	9.95	281.407035	Drive-thru	G
253	NaN	10712	2022-12-29 00:00:00	Fries	3.49	630.372493	Drive-thru	G
254	NaN	10713	2022-12-29 00:00:00	Beverages	2.95	677.966102	Drive-thru	G

255 rows × 10 columns



```
In [6]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 255 entries, 0 to 254
Data columns (total 10 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Unnamed: 0   0 non-null     float64
1   Unnamed: 1   255 non-null   object
2   Unnamed: 2   255 non-null   object
3   Unnamed: 3   255 non-null   object
4   Unnamed: 4   255 non-null   object
5   Unnamed: 5   255 non-null   object
6   Unnamed: 6   255 non-null   object
7   Unnamed: 7   255 non-null   object
8   Unnamed: 8   255 non-null   object
9   Unnamed: 9   255 non-null   object
dtypes: float64(1), object(9)
memory usage: 20.1+ KB
```

1. Delete unwanted/ NaN Columns

```
In [7]: data.drop(columns = "Unnamed: 0", inplace = True)
```

```
In [8]: data.head() #Removed one column
```

```
Out[8]:
```

	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	Unnan
0	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Mani
1	10452	2022-11-07 00:00:00	Fries	3.49	573.065903	Online	Gift Card	Jack
2	10453	2022-11-07 00:00:00	Beverages	2.95	745.762712	Online	Gift Card	P P
3	10454	2022-11-07 00:00:00	Sides & Other	4.99	200.400802	In-store	Gift Card	Joao
4	10455	2022-11-08 00:00:00	Burgers	12.99	569.668976	In-store	Credit Card	W M

2. Change the Column names

```
In [9]: data.loc[0]
```

```
Out[9]: Unnamed: 1      Order ID
        Unnamed: 2      Date
        Unnamed: 3      Product
        Unnamed: 4      Price
        Unnamed: 5      Quantity
        Unnamed: 6      Purchase Type
        Unnamed: 7      Payment Method
        Unnamed: 8      Manager
        Unnamed: 9      City
        Name: 0, dtype: object
```

```
In [10]: data.columns
```

```
Out[10]: Index(['Unnamed: 1', 'Unnamed: 2', 'Unnamed: 3', 'Unnamed: 4', 'Unnamed: 5',
               'Unnamed: 6', 'Unnamed: 7', 'Unnamed: 8', 'Unnamed: 9'],
              dtype='object')
```

```
In [11]: data.columns = data.loc[0]
```

```
In [12]: data.head()
```

```
Out[12]:
```

	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City
0	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City
1	10452	2022-11-07 00:00:00	Fries	3.49	573.065903	Online	Gift Card	Tom Jackson	London
2	10453	2022-11-07 00:00:00	Beverages	2.95	745.762712	Online	Gift Card	Pablo Perez	Madrid
3	10454	2022-11-07 00:00:00	Sides & Other	4.99	200.400802	In-store	Gift Card	Joao Silva	Lisbon
4	10455	2022-11-08 00:00:00	Burgers	12.99	569.668976	In-store	Credit Card	Walter Muller	Berlin

```
In [13]: data.drop(0, inplace = True) #Removed the first row permanently
```

```
In [14]: data.head()
```

Out[14]:

	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City
1	10452	2022-11-07 00:00:00	Fries	3.49	573.065903	Online	Gift Card	Tom Jackson	London
2	10453	2022-11-07 00:00:00	Beverages	2.95	745.762712	Online	Gift Card	Pablo Perez	Madrid
3	10454	2022-11-07 00:00:00	Sides & Other	4.99	200.400802	In-store	Gift Card	Joao Silva	Lisbon
4	10455	2022-11-08 00:00:00	Burgers	12.99	569.668976	In-store	Credit Card	Walter Muller	Berlin
5	10456	2022-11-08 00:00:00	Chicken Sandwiches	9.95	201.005025	In-store	Credit Card	Walter Muller	Berlin

3. Remove extra spaces in Manager column

In [15]: `data['Manager'].unique()`

Out[15]: array(['Tom Jackson', 'Pablo Perez', 'Joao Silva', 'Walter Muller', 'Remy Monet', 'Remy Monet', 'Remy Monet', 'Remy Monet', 'Pablo Perez', 'Pablo Perez', 'Pablo Perez', 'Joao Silva', 'Tom Jackson'], dtype=object)

In [16]: `data['Manager'] = data['Manager'].str.strip().str.replace(r'\s+', ' ', regex=True)`

4. Remove all duplicates

In [17]: `data.describe()`

C:\Users\hmmuz\AppData\Local\Temp\ipykernel_1228\3175101911.py:1: FutureWarning: The behavior of value_counts with object-dtype is deprecated. In a future version, this will *not* perform dtype inference on the resulting index. To retain the old behavior, use `result.index = result.index.infer_objects()`

```
data.describe()
```

Out[17]:

	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	
count	254	254	254	254.00	254.000000	254	254	254	
unique	254	53	5	7.00	29.000000	3	3	5	
top	10452	2022-11-09 00:00:00	Chicken Sandwiches	12.99	200.400802	Online	Credit Card	Tom Jackson	Loi
freq	1	5	52	52.00	49.000000	107	120	75	



No duplicates here, as count and unique are equal!

In [18]: `data[data.duplicated()]`

Out[18]:

	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City
--	----------	------	---------	-------	----------	---------------	----------------	---------	------

In [19]: `data.drop_duplicates(inplace = True)`

5. Converting Datatypes of Columns

In [20]: `data.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 254 entries, 1 to 254
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Order ID        254 non-null    object
1   Date            254 non-null    object
2   Product         254 non-null    object
3   Price           254 non-null    object
4   Quantity        254 non-null    object
5   Purchase Type   254 non-null    object
6   Payment Method  254 non-null    object
7   Manager         254 non-null    object
8   City            254 non-null    object
dtypes: object(9)
memory usage: 18.0+ KB
```

In [21]: `data.Quantity = data.Quantity.astype(float)`In [22]: `data.Date = pd.to_datetime(data.Date)`

Analyzing Data

a). Most Preferred method for Payment?

In [23]: `data.head(2)`

Out[23]:

	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City
1	10452	2022-11-07	Fries	3.49	573.065903	Online	Gift Card	Tom Jackson	London
2	10453	2022-11-07	Beverages	2.95	745.762712	Online	Gift Card	Pablo Perez	Madrid

In [24]: `data['Payment Method'].unique()`

Out[24]: `array([' Gift Card', ' Credit Card', ' Cash'], dtype=object)`

In [25]: `data['Payment Method'].value_counts()`

Out[25]:

```
Payment Method
Credit Card    120
Cash           76
Gift Card      58
Name: count, dtype: int64
```

In [26]: `data['Payment Method'].value_counts(normalize = True)`

Out[26]:

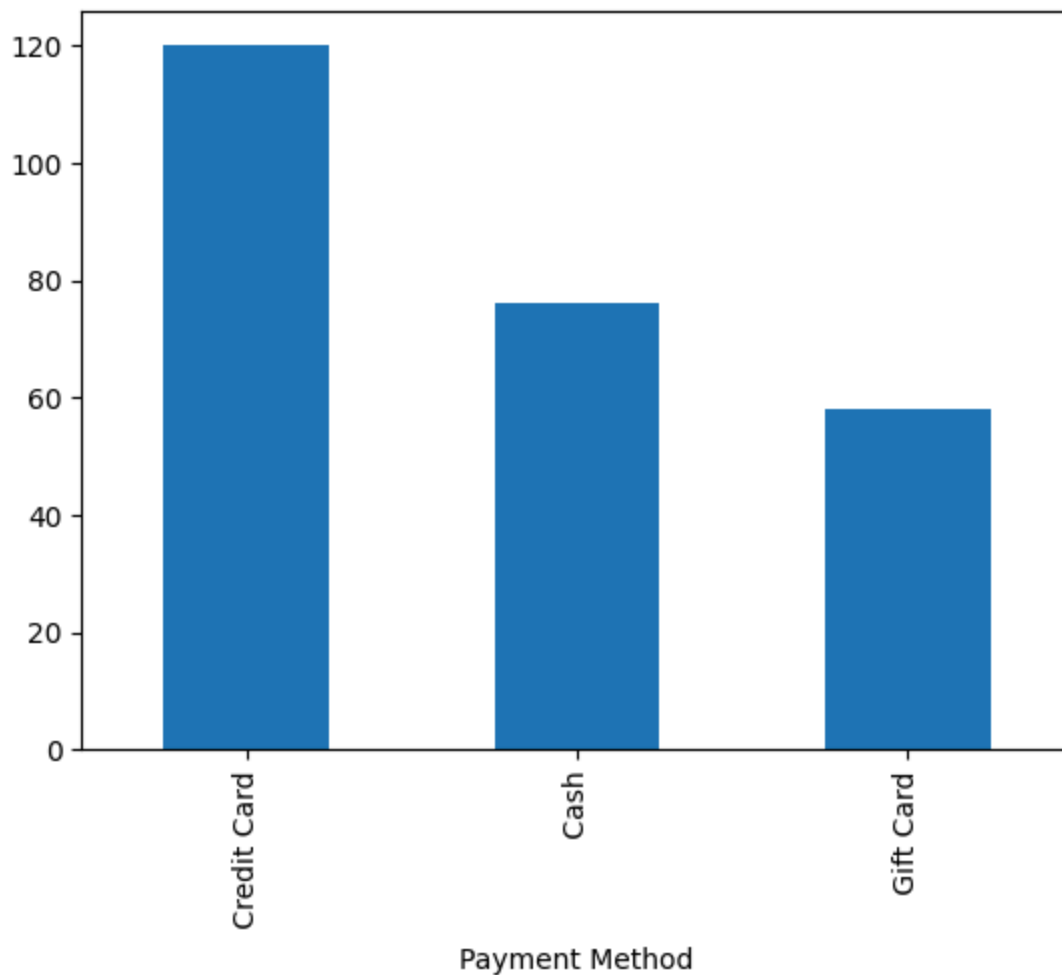
```
Payment Method
Credit Card    0.472441
Cash           0.299213
Gift Card      0.228346
Name: proportion, dtype: float64
```

In [27]: `data['Payment Method'].value_counts(normalize = True)*100` *#to get value in perce*

Out[27]:

```
Payment Method
Credit Card    47.244094
Cash           29.921260
Gift Card      22.834646
Name: proportion, dtype: float64
```

In [28]: `data['Payment Method'].value_counts().plot(kind = 'bar') ;`



b). Most Selling Product?

- by Quantity
- by Revenue

i- By Quantity

In [29]: `data.head(2)`

Out[29]:

	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City
1	10452	2022-11-07	Fries	3.49	573.065903	Online	Gift Card	Tom Jackson	London
2	10453	2022-11-07	Beverages	2.95	745.762712	Online	Gift Card	Pablo Perez	Madrid

In [30]: `data.groupby('Product')['Quantity'].sum()`


```
Out[30]: Product
Beverages      34983.050847
Burgers         29022.324865
Chicken Sandwiches 11135.678392
Fries           32034.383954
Sides & Other   9819.639279
Name: Quantity, dtype: float64
```

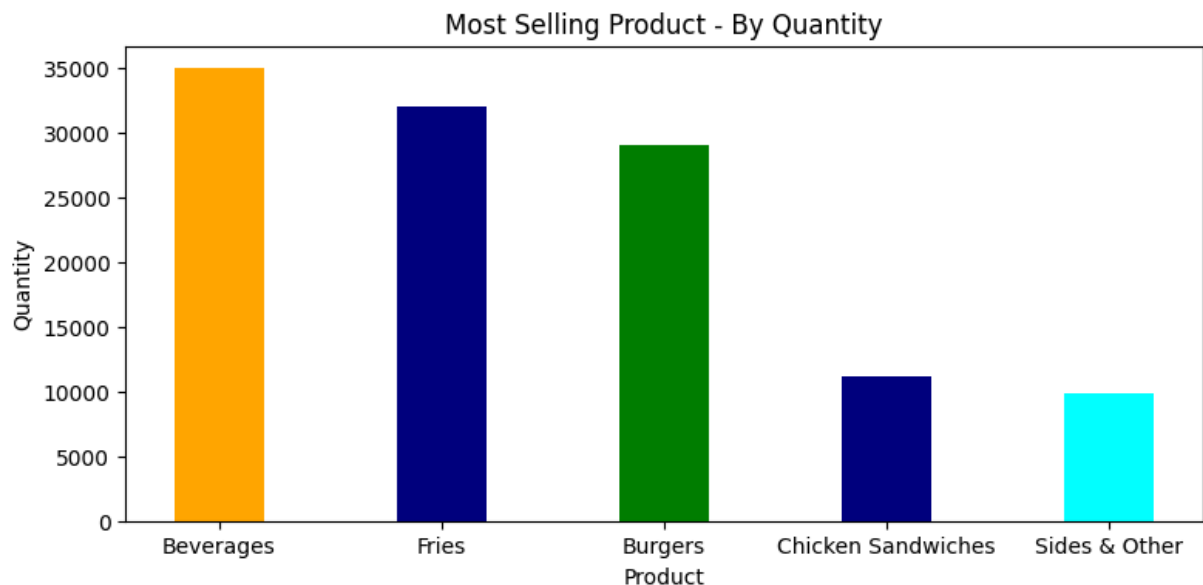
```
In [31]: most_quantity = data.groupby('Product')['Quantity'].sum().sort_values(ascending= Fa
```

```
In [34]: most_quantity = most_quantity.reset_index()
most_quantity
```

```
Out[34]:
```

	index	Product	Quantity
0	0	Beverages	34983.050847
1	1	Fries	32034.383954
2	2	Burgers	29022.324865
3	3	Chicken Sandwiches	11135.678392
4	4	Sides & Other	9819.639279

```
In [45]: plt.figure(figsize = (9,4))
plt.bar(most_quantity['Product'], most_quantity['Quantity'], color =['orange', 'nav
plt.title('Most Selling Product - By Quantity')
plt.xlabel('Product')
plt.ylabel('Quantity');
```



ii- By Revenue

```
In [47]: data['Revenue'] = data['Quantity']*data['Price']
```

```
In [48]: data.head(2)
```

Out[48]:

	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City	Revenue
1	10452	2022-11-07	Fries	3.49	573.065903	Online	Gift Card	Tom Jackson	London	2000.0
2	10453	2022-11-07	Beverages	2.95	745.762712	Online	Gift Card	Pablo Perez	Madrid	2200.0

In [54]: `data.groupby("Product")["Revenue"].sum().sort_values(ascending=False)`

Out[54]:

Product	Revenue
Burgers	377000.0
Fries	125674.498567
Chicken Sandwiches	114639.19598
Beverages	103200.0
Sides & Other	49000.0

Name: Revenue, dtype: object

In [62]: `most_revenue = data.groupby('Product')['Revenue'].sum().sort_values(ascending=False)`
`most_revenue`

Out[62]:

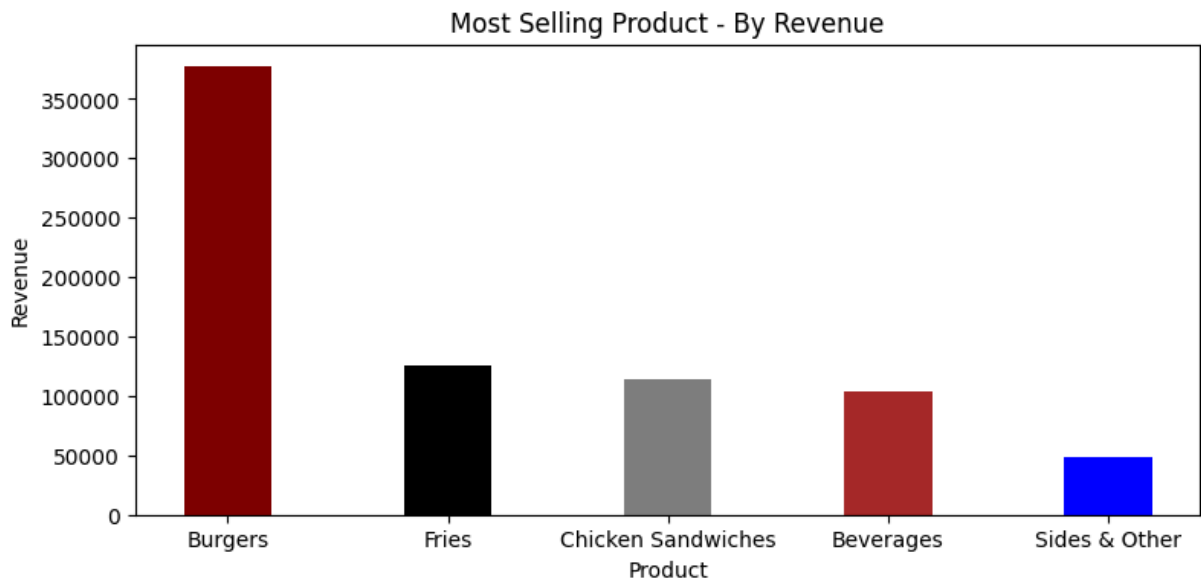
Product	Revenue
Burgers	377000.0
Fries	125674.498567
Chicken Sandwiches	114639.19598
Beverages	103200.0
Sides & Other	49000.0

Name: Revenue, dtype: object

In [66]: `most_revenue = most_revenue.reset_index()`

In [74]:

```
plt.figure(figsize = (9,4))
plt.bar(most_revenue['Product'], most_revenue['Revenue'], color = ['maroon', 'black'])
plt.title('Most Selling Product - By Revenue')
plt.xlabel('Product')
plt.ylabel('Revenue');
```



c). Which Manager earned maximum Revenue?

```
In [78]: data.Manager.unique()
```

```
Out[78]: array(['Tom Jackson', 'Pablo Perez', 'Joao Silva', 'Walter Muller',
                'Remy Monet'], dtype=object)
```

```
In [79]: data.groupby('Manager')['Revenue'].sum().sort_values(ascending = False)
```

```
Out[79]: Manager
Joao Silva      241713.694547
Tom Jackson      211200.0
Pablo Perez      136200.0
Walter Muller     100600.0
Remy Monet        79800.0
Name: Revenue, dtype: object
```

d). Date wise Revenue?

```
In [80]: data.head(2)
```

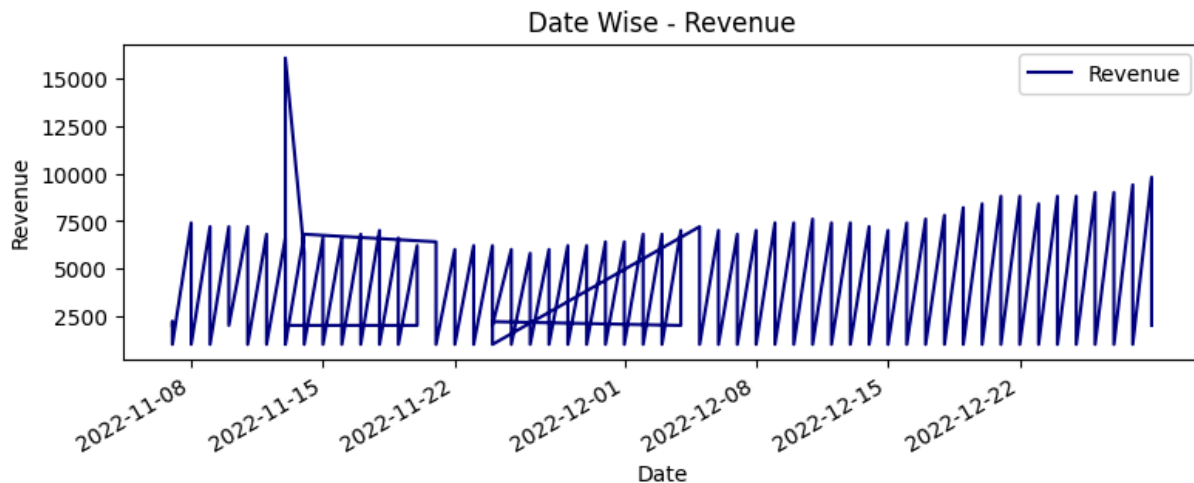
```
Out[80]:
```

	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City	Revenue
1	10452	2022-11-07	Fries	3.49	573.065903	Online	Gift Card	Tom Jackson	London	2000.00
2	10453	2022-11-07	Beverages	2.95	745.762712	Online	Gift Card	Pablo Perez	Madrid	2200.00

```
In [84]: data['Date'].dtype
```

```
Out[84]: dtype('<M8[ns]')
```

```
In [89]: data.plot('Date', 'Revenue', color=['navy'], figsize=(9,3))
plt.title('Date Wise - Revenue')
plt.xlabel('Date')
plt.ylabel('Revenue');
```



e). What is the average Revenue of November and December?

```
In [90]: data['Month'] = data['Date'].dt.month
```

```
In [91]: data.head(2)
```

```
Out[91]:
```

	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City	Revenue
1	10452	2022-11-07	Fries	3.49	573.065903	Online	Gift Card	Tom Jackson	London	2000
2	10453	2022-11-07	Beverages	2.95	745.762712	Online	Gift Card	Pablo Perez	Madrid	2200

```
In [96]: m11 = data[data['Month'] == 11]
```

```
In [97]: m11
```

Out[97]:

	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City
1	10452	2022-11-07	Fries	3.49	573.065903	Online	Gift Card	Tom Jackson	London
2	10453	2022-11-07	Beverages	2.95	745.762712	Online	Gift Card	Pablo Perez	Madrid
3	10454	2022-11-07	Sides & Other	4.99	200.400802	In-store	Gift Card	Joao Silva	Lisbon
4	10455	2022-11-08	Burgers	12.99	569.668976	In-store	Credit Card	Walter Muller	Berlin
5	10456	2022-11-08	Chicken Sandwiches	9.95	201.005025	In-store	Credit Card	Walter Muller	Berlin
...	...	...	...	...	...	...	...	...	...
109	10568	2022-11-30	Beverages	2.95	677.966102	Online	Credit Card	Tom Jackson	London
110	10569	2022-11-30	Sides & Other	4.99	200.400802	Online	Credit Card	Tom Jackson	London
128	10537	2022-11-24	Fries	3.49	630.372493	Drive-thru	Credit Card	Pablo Perez	Madrid
129	10538	2022-11-24	Beverages	2.95	745.762712	Drive-thru	Credit Card	Pablo Perez	Madrid
130	10539	2022-11-24	Sides & Other	4.99	200.400802	Drive-thru	Credit Card	Pablo Perez	Madrid

113 rows × 11 columns



```
In [98]: m11.Revenue.mean()
```

Out[98]: np.float64(2939.0592437808386)

```
In [99]: m12 = data[data['Month'] == 12]
```

```
In [100... m12
```

Out[100...

	Order ID	Date	Product	Price	Quantity	Purchase Type	Payment Method	Manager	City
111	10570	2022-12-01	Burgers	12.99	492.686682	Online	Credit Card	Tom Jackson	London
112	10571	2022-12-01	Chicken Sandwiches	9.95	201.005025	Online	Credit Card	Tom Jackson	London
113	10572	2022-12-01	Fries	3.49	573.065903	Online	Credit Card	Remy Monet	Paris
114	10573	2022-12-01	Beverages	2.95	677.966102	Online	Credit Card	Remy Monet	Paris
115	10574	2022-12-01	Sides & Other	4.99	200.400802	Online	Credit Card	Remy Monet	Paris
...	...	...	...	...	...	...	...	...	...
250	10709	2022-12-28	Sides & Other	4.99	200.400802	Drive-thru	Gift Card	Walter Muller	Berlin
251	10710	2022-12-29	Burgers	12.99	754.426482	Drive-thru	Gift Card	Walter Muller	Berlin
252	10711	2022-12-29	Chicken Sandwiches	9.95	281.407035	Drive-thru	Gift Card	Walter Muller	Berlin
253	10712	2022-12-29	Fries	3.49	630.372493	Drive-thru	Gift Card	Walter Muller	Berlin
254	10713	2022-12-29	Beverages	2.95	677.966102	Drive-thru	Gift Card	Walter Muller	Berlin

141 rows × 11 columns



In [101...

m12.Revenue.mean()

Out[101...

np.float64(3102.127659574468)

f). What is Standard Deviation for Revenue & Quantity?

In [103...

data['Quantity'].std()

Out[103...

np.float64(214.88955129185314)

In [104...

data['Revenue'].std()

Out[104...

np.float64(2420.1241927718042)

g). What is Variance for Revenue & Quantity?

In [105... `data['Revenue'].var()`

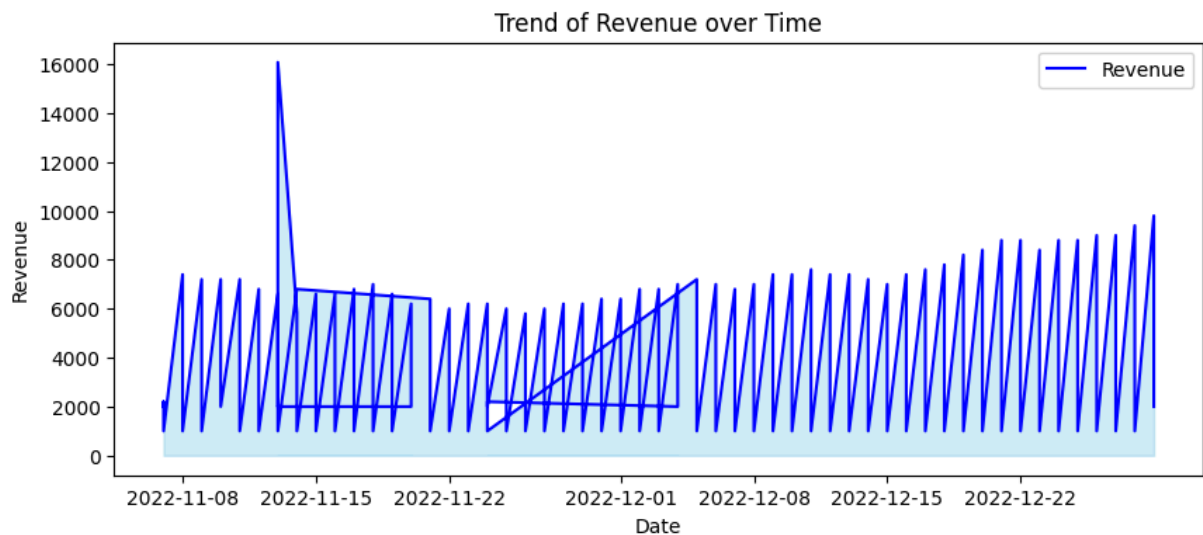
Out[105... `np.float64(5857001.108439377)`

In [106... `data['Quantity'].var()`

Out[106... `np.float64(46177.51925441398)`

h). Is Revenue Increasing or Decreasing over time?

```
In [113... import matplotlib.pyplot as plt
data['Date'] = pd.to_datetime(data['Date'])
data['Revenue'] = pd.to_numeric(data['Revenue'], errors='coerce')
plt.figure(figsize = (10, 4))
plt.plot(data['Date'], data['Revenue'], color='blue', label='Revenue')
plt.fill_between(data['Date'], data['Revenue'], color='skyblue', alpha=0.4)
plt.title('Trend of Revenue over Time')
plt.xlabel('Date')
plt.ylabel('Revenue')
plt.legend()
plt.show()
```



h). Average Quantity sold and Average Revenue for each Product?

```
In [116... data.groupby('Product')[['Revenue', 'Quantity']].agg({'Revenue' : 'mean', 'Quantity'
```

Out[116...

	Revenue	Quantity
Product		
Beverages	2064.000000	699.661017
Burgers	7250.000000	558.121632
Chicken Sandwiches	2204.599923	214.147661
Fries	2464.205854	628.125176
Sides & Other	1000.000000	200.400802

In []: